Experiment 5 – Laser diodes

Explanation: goal, principle, procedure and why we need it later

1. The experiment in one sentence

A laser diode is an LED with a resonator: we measure the current (threshold) above which spontaneous LED light turns into amplified, narrow-band laser light, how this depends on temperature, and what the beam looks like (near and far field).

2. Goal of the experiment

- Record P-I curves at 20, 30, 40 °C; determine the threshold current I_{th}, the slope dP/dI and the differential quantum efficiency.
- Determine T_0 and $I_{th,0}$ from $I_{th}(T)$.
- Record spectra below and above threshold (20–40 °C) and determine $d\lambda/dT$.
- Take near- and far-field pictures with the camera, estimate the facet width and explain the differences.

3. Principle – the logic behind it

3.1 From LED to laser

In a laser diode the pn junction is pumped so strongly in forward direction that there are more electrons in the conduction band than in the valence band (at the same energy) – **population inversion**. Existing photons then trigger further recombinations with the same energy, phase and direction (**stimulated emission**). The cleaved crystal facets form a **Fabry-Perot resonator**; with $n \approx 3.5$ they reflect only $R = ((3.5-1)/(3.5+1))^2 = 31\ \%$ – the losses are high, hence the high threshold current density.

3.2 Structure and threshold

As for the LED a double heterostructure is used (today mostly a quantum well with a separate-confinement waveguide, SCH); the waveguide guides the light in the active zone (confinement factor $\Gamma \leq 1$). **Threshold:** gain = losses, $g_{th} = \alpha_0 + (1/2L)\ \ln(1/R_1R_2)$. Low threshold current density: thin active layer, high Γ, long wavelength, long resonator ($\geq \approx 100\ \mu m$), high reflectivity.

3.3 The P-I curve

Below I_{th} spontaneous emission dominates (LED-like, little power); above it the power rises linearly: $P = \frac{1}{2}\ \eta_i\ \eta_e\ (h\nu/q)(I - I_{th})$. The slope gives the differential quantum efficiency $\eta_d = (dP/dI)\cdot q/(h\nu)$ (660 nm: $h\nu/q = 1.88$ V). With temperature the threshold current rises, $I_{th} = I_{th,0}\ e^{T/T0}$, and the slope falls: non-radiative processes (phonon scattering, Auger) increase. A large T_0 (50–200 °C) means low temperature sensitivity.

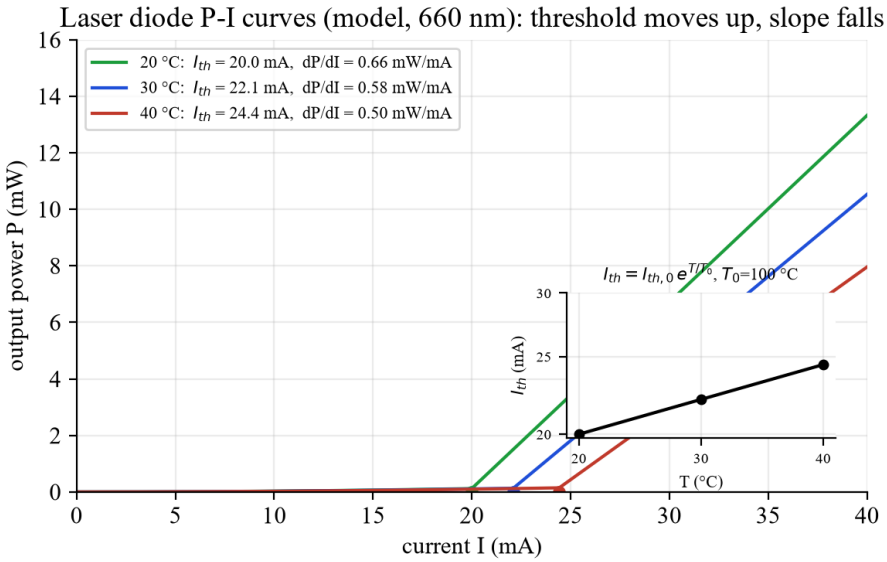


Fig. 1: P-I curves (model): the threshold moves up, the slope falls.

3.4 Spectrum and wavelength

Below threshold: broad spectrum (≈ 10–20 nm). Above: a few narrow longitudinal modes spaced by $\Delta\lambda = \lambda^2/(2nL)$ (≈ 0.2 nm for L = 300 µm). With rising temperature the gain maximum (band gap ↓) moves ≈ 0.2–0.3 nm/K towards red, the modes only ≈ 0.06 nm/K – hence **mode hops**. Current → heating → similar effect.

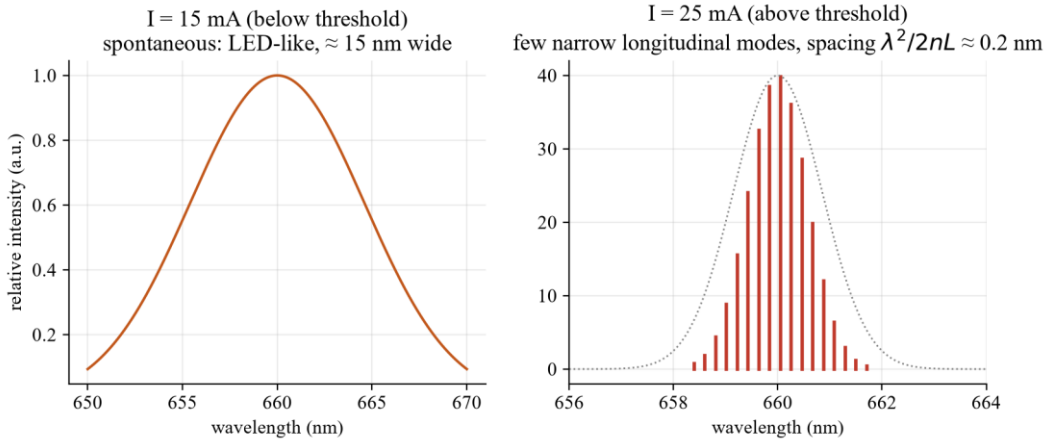


Fig. 2: Spectrum below and above threshold (schematic).

3.5 Beam profile

The waveguide is thin (≈ 1 µm) and wide (5–200 µm): the **near field** (image of the facet) is small and elongated, the **far field** (Fourier transform) is elliptical and rotated by 90°: diffraction angle $\Theta = \lambda/(\pi d)$ – perpendicular to the layers ≈ 90°, lateral ≈ 15°. Collimating optics are therefore needed. Near field: microscope (M = b/f − 1, f = 6 mm), scale via a mm grid.

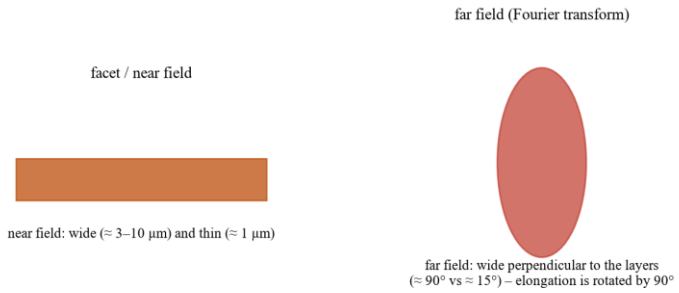


Fig. 3: Near field and far field (schematic).

4. Procedure step by step

Task	What to do?	Evaluation
1 P-I	Set T = 20/30/40 °C (sensor resistance in kΩ), wait; current in 1 mA steps (0.5 mA at the knee) up to 2.5 mW	Line above threshold: I_{th}, dP/dI, η_d; ln $I_{th}(T) \rightarrow T_0$, $I_{th,0}$
2 Spectrum	15 mA and 25 mA, 20–40 °C in 5 °C steps; dark spectrum; avoid saturation (check the threshold at 40 °C!)	Overlay spectra, $\lambda(T) \rightarrow d\lambda/dT$
3 Near/far field	Camera, gray filters; current below → above threshold; mm grid; measure b; then far-field lens	M = b/f − 1, facet width = near-field width/M; describe the far field

Optional: LTspice model of the P-I curves at 3 temperatures: Laserdiode_PI_Temperatur.asc (or *04_LTspice/LTspice_starten.bat*).

5. What to watch in the lab

- **Laser safety:** never look into the beam (even 2.5 mW can damage the retina); gray filters before camera/detectors; do not overload the spectrometer CCD.
- **Raise the current slowly** (constant current, knob fully left first); laser action from ≈ 20 mA; stop at 2.5 mW.
- **Wait until the temperature is stable** (calibration kΩ → °C); otherwise the data belong to a wrong temperature.
- Determine the threshold by extrapolating the straight line, not by eye.
- Protocol: expectation – measurement – error analysis for every task.

6. Why do we need this later?

- **Experiment 6 (DFB laser):** the same FP laser has several modes and mode hops; the DFB laser selects exactly one mode with a grating – temperature tuning becomes continuous.
- **Experiment 4:** threshold, P-I curve and spectrum explain the difference between LED and laser.
- **Experiment 1:** laser diodes are the transmitters in fibre systems (1.3/1.55 µm); temperature control and beam shaping are decisive there.
- Practice: laser pointers, bar-code readers, data links, laser printers, pumping solid-state lasers.

7. Typical questions – short answers

Question	Short answer
LED vs. laser diode?	Laser: inversion + resonator → stimulated emission, narrow, directed, linear P-I above threshold.
Reflection GaAs-air?	$R = ((3.5-1)/(3.5+1))^2 \approx 31$ %.
Threshold condition?	Gain = losses: $g_{th} = \alpha_0 + (1/2L)\ \ln(1/R_1R_2)$.
What lowers the threshold?	Thin active layer, high Γ, long resonator, high reflectivity, low temperature.

How to get η_d?	$\eta_d = (dP/dI)\cdot q/(h\nu)$ from the slope above threshold.
What is T_0?	Characteristic temperature in $I_{th} = I_{th,0}\, e^{T/T0}$; large = insensitive.
$\lambda(T)$?	≈ 0.2–0.3 nm/K (gain maximum), mode hops.
Mode spacing?	$\Delta\lambda = \lambda^2/(2nL)$, $\Delta\nu = c/(2nL)$.
Why is the beam elliptical?	Thin waveguide → strong diffraction ($\Theta = \lambda/(\pi d)$) perpendicular, weak lateral.
Near vs. far field?	Near field = image of the facet, far field = Fourier transform (rotated by 90°).
Facet width from the image?	$M = b/f - 1$; width = intermediate-image width (from the mm grid)/M.
Safety?	Never look into the beam, gray filters, constant current, raise slowly.

Date: **2 Dec 2026**, pre-test (Antestat) 14:00, building BC, room BC 119 (group 9).